

Joint Learning by Doing

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Abstract

We study how social interactions affect the adoption of a household technology whose benefits are not directly observable to peers, and how joint technology use in turn affects social interactions. We cross-randomize a three-month cookstove trial with regular meetings between randomly assigned women, generating four pair compositions: neither woman has access to the technology, only the respondent does, only her peer does, or both do. Technology access alone increases usage and knowledge and shifts beliefs about its benefits by 0.35 standard deviations. However, when both women have access to the technology, they have less favorable beliefs, even though they attend significantly more meetings. Their attention is also more concentrated on specific dimensions of the technology. Survey evidence and a vignette experiment suggest that social costs may reduce information seeking: joint access increases expected judgement when asking for help, while close friendship increases willingness to ask. At the same time, joint technology use has substantial effects on social interactions. It increases engagement between randomly matched peers, closes the cross-caste gap in interaction, and reduces loneliness.

Keywords: technology adoption, learning-by-doing, social networks, field experiment.

JEL codes: O12, O33, D83, D85, C93, I12, Z13.

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1 Introduction

Technology adoption is often slow even when new technologies are accessible and affordable (16; 8). Social interactions can facilitate adoption by allowing individuals to learn from the experiences of others (11; 1; 3; 4; 7). Learning from others can occur either through information that is available without exerting much effort (such as observing how well a farmer’s crop performs), or through information that must be actively elicited from peers (such as asking whether a new fertilizer was worth using). The second channel is particularly important for technologies where peer outcomes are not easily observed. However, eliciting information can itself be costly: asking questions can reveal ignorance and affect reputation (9).

This paper studies how distinct channels of social learning affect the adoption of a private-use experience good. We also study the reverse relationship: whether access to a technology can itself affect social interactions. Joint access to a technology may create a shared purpose that increases social engagement, including among individuals who would otherwise not interact as much. To answer both questions, we implement a field experiment with 1,555 women across 40 villages in rural Nepal where we cross-randomize access to a temporary trial of a technology and one-to-one meetings with a randomly chosen peer. The technology is an electric induction cookstove, a private-use experience good whose use is largely opaque to neighbors. We eliminate selection by assigning exogenously chosen peers. The cross-randomization generates four pair compositions: neither peer has the technology, the respondent has it but the peer does not, the peer has it but the respondent does not, or both have it. This allows us to compare learning from signals generated privately through own use with learning from signals transmitted by a peer or generated jointly through shared use.

Several features of our context make this an informative setting to answer these questions. Firewood is the primary cooking fuel for 52% of households and imposes substantial health costs: among firewood users, 85% report eye irritation from smoke, 63% report coughing or choking, and 63% report headaches. While 48% of households use LPG as their primary fuel, only 7% of respondents know how to operate an electric cookstove. 86% report they would be more likely to adopt if a friend adopts first, most often because a friend’s adoption would encourage them to use the stove (57%) or make them confident that it works. At the same time, 76% rarely or never ask a friend for advice on a kitchen matter such as buying or using an appliance.

Women in our study are allocated to one of four intervention arms: control, technology trial only, peer exposure only, or technology and peer exposure. Respondents in the trial group are informed at the outset that they will have to return it at the end of the three-month period, but that they will be offered the option to purchase it. Half of technology recipients are additionally randomized to an air-quality demonstration, in which they are shown the

reading from an air-quality monitor while cooking with firewood and again while cooking with the electric stove. Respondents in the peer exposure group are invited to share meals with a randomly chosen peer in either their or their partner’s house. They are asked to meet every two weeks over three months. The first meeting is enumerator-verified and subsequent meetings are incentivized but unsupervised. We collect data in a three-month endline as well as a six-month follow-up with a subset of participants. A one-year follow-up is due to begin in November, 2026.

First, we find that access to the technology increases adoption. Relative to within-village controls, technology access alone increases incentivized willingness to pay for it, expenditure on household electricity, and the probability of a wiring upgrade, which is a complementary investment. It also increases self-reported three-month use of the stove by 66 percentage points. However, exposure to a peer has no statistically detectable effect on the harder measures of adoption even though self-reported use is 11 percentage points higher in the combined arm than under technology access alone ($p = 0.03$).

The absence of effects on most adoption measures cannot be attributed to low meeting participation. Relative to pairs in which neither partner receives the technology, pairs where both receive the technology (“both-stove”) attend 1.8 additional meetings, are 53 percentage points more likely to cook together, and are 19 percentage points more likely to meet outside the scheduled sessions. Next, we find that “both-stove” treatment increases respondents’ confidence in their ability to operate the stove, but does not increase their knowledge of how to operate it. Respondents in both-stove pairs are 13 percentage points more likely to report being confident or very confident in operating the stove than respondents with technology access alone ($p = 0.047$). Despite this engagement and higher confidence, they do not achieve a significantly higher score on the enumerator-administered operating checklist, which measures knowledge of how to operate the stove.

Technology access alone also shifts the belief index, which captures how favourably respondents assess the stove’s expected benefits (health, cooking time, traditional foods, indoor air quality) and practical risks (electricity supply and bills, breakage, shocks), by 0.35σ . However, both-stove respondents have significantly less favorable beliefs than respondents with technology access alone, with a belief index that is 0.34σ lower ($p = 0.03$). Their beliefs are not statistically different from those of pairs in which neither partner has access. By contrast, respondents who meet a peer with access but do not have the technology themselves have more favorable beliefs than pairs in which neither partner has access ($p = 0.063$), consistent with observational learning from the peer’s experience.

To understand these effects on beliefs, we next examine how the treatments affect the allocation of attention to different dimensions. We find that joint access does not significantly affect the number of cooking dimensions to which respondents pay attention, but it does concentrate their attention across categories. Relative to technology access alone, both-stove

respondents have more concentrated attention under all three measures we construct, with the differences significant at the 10-percent level for two of the three measures. Consistent with this, both-stove pairs are also 17 percentage points less likely than pairs with no technology to rank “asking questions and getting help” among the three most helpful aspects of the meetings ($p < 0.10$).

In a separate vignette experiment, we find evidence of high predicted embarrassment and judgement in asking questions among pairs where both have the technology. These estimates are consistent with joint ownership increasing the perceived social cost of eliciting information. In the same experiment, describing the peer as a close friend significantly increases willingness to ask for help. Consistent with familiarity lowering the cost of asking, women who already knew their assigned peer engage more with their peer. Among women for whom both partners had access to the technology, those who knew their peer *ex ante* use the cookstove more frequently, though their beliefs and willingness to pay are no different.

We rationalize these results using a model in which joint use raises ambient information available from a peer but also raises the cost of eliciting signals, and can concentrate attention on particular dimensions, so that more contact need not produce more learning. Randomizing the air-quality demonstration provides a test of this mechanism. If the learning penalty arises because asking reveals ignorance, it should disappear when the peer holds information the respondent could not have been expected to know herself. We find that, among technology recipients, receiving an air quality reading oneself has no effect on outcomes. However, relative to the technology-only arm, meeting a stove-owning peer (“both-stoves”) who did not receive the reading reduces the belief index by 0.54σ and also reduces the knowledge index, although the latter effect is not statistically significant. In contrast, when the peer received the reading, both gaps are close to zero, and the pair is 22 percentage points more likely to have discussed the cookstove. The difference across the two joint-access cells is significant at the 10-percent level for both beliefs and knowledge. The pooled joint-access belief penalty of 0.34σ is therefore concentrated entirely among pairs whose peer held no information the respondent could not have been expected to hold herself.

Although joint use does not raise knowledge or make beliefs more favorable beyond own access, joint use of the technology has positive effects on social interaction. Both-stove pairs report 0.30σ lower loneliness than control ($p < 0.05$), including a 21-percentage-point reduction in lacking companionship relative to a control mean of 42 percent. Individuals who received the stove and were assigned to peer meetings also have significantly higher social interactions than the meeting-only arm ($p = 0.04$), measured using an index that combines house visits, shared meals, and the numbers of friends, advice contacts, and borrowing or lending contacts. Finally, joint access to the technology reduces the cross-caste gap in engagement: cross-caste pairs attend significantly fewer of the six sessions than same-caste pairs ($p = 0.002$) and are 20 percentage points less likely to meet outside the scheduled sessions. However, when both partners have the technology these gaps close: among cross-

caste pairs, joint access raises attendance by 3.1 sessions and the probability of meeting outside the sessions by 43 percentage points (both $p < 0.001$). Further, six months later, joint-access pairs report significantly more frequent contact with their assigned peer than pairs in which neither partner has the technology ($p = 0.05$), and are 22 percentage points more likely to be in touch weekly or more often ($p < 0.01$).

Related Literature: We make two contributions to the existing literature. First, we contribute to research on social learning and technology adoption (14; 13; 15; 1; 11; 18; 4; 6; 3; 7; 16; 19), which studies how individuals learn from peers’ choices, outcomes, and information. We study social learning for a private-use technology by experimentally separating learning by doing, observational learning, information exchange, and joint learning by doing. We also build on (9), who show that seeking information can be socially costly, and (8), who show how reputational concerns can affect information transmission. Our results suggest that more interaction can increase confidence without increasing knowledge, can concentrate attention, and need not translate into more favorable beliefs or greater adoption. The social cost of seeking information is one plausible reason.

Second, we contribute to a growing literature on the formation of social networks and how policy interventions affect network interactions (e.g., 2; Fleischman). Existing literature on technology adoption has largely treated networks as a primitive (e.g., 11; 3; 4). When an intervention changes the network itself, ignoring these network changes can lead to biased estimates of treatment effects (10). We show that a jointly used *technology* can also encourage network interactions: joint use reduces loneliness, and reduces the cross-caste gap in engagement. In doing so, we relate the two strands of literature on technology-adoption and network-formation by showing that encouraging joint use of a technology can potentially sustain social interaction. Our findings are particularly relevant for caste-segregated settings, where cross-group interactions are difficult to engineer through standard contact interventions. For example, (17) shows that collaborative contact, in which individuals share a task, reduces cross-caste barriers while adversarial contact does not. We show that joint access to a technology can incentivize cross-group contact.

The rest of the paper proceeds as follows. Section 2 describes the setting and the experimental design. Section 3 provides a conceptual framework that rationalises the results. Section 4 reports effects on technology adoption and mechanisms underlying the observed effects. Section 5 reports effects on social interactions three months and six months after the intervention.

2 Context and Design

Our study covers 40 villages in rural Nepal and 1,555 women, all of whom completed a baseline survey before randomization. Table 1 reports their demographic characteristics.

The median respondent is 43 years old and lives in a household of four with a monthly income of NPR 20,000 (about USD 130). Agriculture is the main source of income for 59 percent of households, remittances for 13 percent and government employment for 7 percent. Schooling is limited: 51 percent have at most primary education and a further 32 percent completed grades 6–10. In 53 percent of households, at least one member has migrated out of the village. Ninety percent of respondents do most of the household cooking, so they are the relevant decision makers for the technology we study. Caste composition is concentrated: 56 percent of respondents are Tamang and 23 percent Chettri, with Dalit, Newar, Brahmin and Rai households making up the remainder.

Table 1: Sample Characteristics at Baseline

	Mean	SD	N
<i>Panel A. Demographics and household</i>			
Age (years)	43.9	13.6	1,555
Household size	4.0	1.7	1,555
At least one household member has migrated out of the village	0.527	0.499	1,489
Respondent does most of the household cooking	0.898	0.302	1,555
<i>Panel B. Household income</i>			
Monthly household income (NPR)	22768.6	18528.3	1,555
Main income source: agriculture	0.588	0.492	1,555
Main income source: remittances	0.133	0.340	1,555
Main income source: government employment	0.070	0.255	1,555
Main income source: shop	0.051	0.220	1,555
Main income source: trade	0.047	0.212	1,555
Main income source: agro-business	0.047	0.212	1,555
Main income source: other	0.081	0.273	1,555
<i>Panel C. Education</i>			
Schooling: none, informal, or primary	0.508	0.500	1,555
Schooling: grades 6–10	0.322	0.467	1,555
Schooling: grades 11–12	0.121	0.326	1,555
Schooling: university	0.049	0.216	1,555
<i>Panel D. Caste</i>			
Caste: Tamang	0.563	0.496	1,555
Caste: Chettri	0.232	0.422	1,555
Caste: Dalit	0.065	0.247	1,555
Caste: Newar	0.063	0.243	1,555
Caste: Brahmin	0.044	0.205	1,555
Caste: Rai or other	0.034	0.182	1,555

Notes: All 1,555 women in the experimental sample, measured at baseline.

2.1 Social Interaction and Networks

We measure respondent social networks in the baseline survey. Table 2 reports the baseline measures. Respondents name 3 people in the village as friends i.e. people they borrow from or lend to, exchange advice with, or would contact in an emergency. 40 percent report lacking companionship sometimes or often. In the month before the survey, 52 percent of respondents rarely or never visited or were visited by another household in the village, and 63 percent rarely or never shared a meal or tea with someone in their own or another home. This affects their ability to observe the use of a hidden technology. Moreover, 76 percent rarely or never asked a friend for advice on a kitchen-related topic such as buying an appliance, using cooking equipment or trying a new method.

Table 3 shows that women who are more socially connected are more likely to seek household-technology advice. They also report greater confidence in using and troubleshooting the technology and are more likely to know how to operate it. Moreover, more socially connected women report substantially lower loneliness. These correlations are consistent with social isolation imposing costs not only in terms of well-being, but also in terms of access to information and opportunities to learn from others.

Table 2: Social Interaction and Networks at Baseline

	Mean	SD	N
<i>Panel A. Frequency of interaction (share rarely or never)</i>			
Visited or was visited by other households rarely or never	0.515	0.500	1,555
Shared a meal or tea rarely or never	0.630	0.483	1,555
Cooked a meal together with someone else rarely or never	0.683	0.465	1,487
Sought household-technology-related advice rarely or never	0.756	0.430	1,555
<i>Panel B. Network size</i>			
Number of friends named	3.0	0.7	1,516
Number of friends named who are not family	2.5	1.1	1,516
Number of households visited in the last month	2.3	1.0	1,511
<i>Panel C. Loneliness and openness to strangers</i>			
Lacks companionship sometimes or often	0.400	0.490	1,555
Willing (4 or 5 of 5) to share a meal with a randomly chosen woman	0.745	0.436	1,555
Comfortable (4 or 5 of 5) hosting a randomly chosen woman	0.648	0.478	1,555

2.2 The Technology

The technology we focus on is an electric induction cookstove. It is a private-use experience good as its benefits are realized through continued use and are largely invisible to neighbors. It is also close to new in this setting. Only 7 percent of respondents know how to operate

Table 3: Social Connectedness, Well-Being, and Learning at Baseline

	Loneliness	Household-technology advice	Technology-use and troubleshooting confidence	Knows how to operate the technology
Respondent-reported connectedness index	-0.130*** (0.028)	0.281*** (0.029)	0.197*** (0.028)	0.025*** (0.007)
Connectedness p-value	0.000	0.000	0.000	0.001
Village fixed effects	Yes	Yes	Yes	Yes
Observations	1508	1508	1508	1503
R^2	0.094	0.108	0.071	0.066

an electric cookstove at baseline, firewood is the primary cooking fuel for 52 percent of households and LPG for 48 percent. 77 percent use firewood as a primary or secondary fuel.

Health is a primary reason why a cleaner technology could improve outcomes. Half of respondents rate wood smoke as harmful to health, giving it a 4 or 5 on a five-point scale. Among firewood users, 85 percent report tearing eyes from smoke, 63 percent report coughing or choking and 63 percent report headaches. 10 percent of all households have a member with a respiratory illness. The leading perceived disadvantage of the technology is the electricity supply, listed by 74 percent, followed by costs as well as difficulty of learning how to use it at 29 percent each, and device fragility at 24 percent.¹ The leading perceived advantages include saving time on cooking, named by 75 percent, and health benefits, named by 59 percent.

Latent demand for the technology is high. We first elicit willingness to pay for a stove at baseline using an unincentivized price ladder running up until NPR 5,000 (about USD 33). The mean willingness to pay is NPR 3,108 (USD 20) and the median NPR 3,000 (USD 20). 23 percent accept the highest price and only 3 percent reject every price. Compared to their income, 78 percent would pay at least a tenth of their monthly income for a stove.

The role of peers in increasing adoption: Respondents also expect peers to matter for adoption: 86 percent say they would be likely or very likely to adopt if a friend adopted first. When asked why, the most common reason is motivation: 57 percent say a friend would encourage them to use it. Belief updating comes next: about half say a friend’s adoption would make them confident the technology works. Direct learning is less commonly reported, with only under a third expecting a friend to teach them. At the same time, few peers are available to learn from at baseline in the first place. Of their ten closest friends, respondents think, on average, about 1 has ever used an electric cookstove.

¹Running an induction stove requires reliable electrical power: 96 percent report frequent electricity outages at baseline and adoption could require a wiring upgrade.

Table 4: Cooking, Fuel, Health and Perceptions at Baseline

	Mean	SD	N
<i>Panel A. Cooking and fuel use</i>			
Primary cooking fuel: firewood	0.515	0.500	1,555
Primary cooking fuel: gas cylinder (LPG)	0.482	0.500	1,555
Uses firewood as primary or secondary fuel	0.767	0.423	1,555
Share of weekly cooking done on primary fuel (percent)	82.0	17.8	1,555
Knows how to operate an electric cookstove	0.069	0.254	1,550
<i>Panel B. Smoke and health</i>			
Woodsmoke rated harmful to health (4 or 5 of 5)	0.508	0.500	1,555
Tearing eyes from smoke (firewood users)	0.850	0.357	801
Coughing or choking from smoke (firewood users)	0.633	0.482	801
Headaches from smoke (firewood users)	0.630	0.483	801
Backaches (firewood users)	0.154	0.361	801
Someone in the household suffers from respiratory illness	0.096	0.294	1,555
<i>Panel C. Perceived advantages</i>			
Advantage: saves time	0.749	0.434	1,555
Advantage: health benefits	0.587	0.493	1,555
Advantage: environmental benefits	0.239	0.426	1,555
Advantage: more convenient	0.239	0.427	1,555
<i>Panel D. Perceived drawbacks</i>			
Drawback: depends on electricity	0.736	0.441	1,555
Drawback: expensive relative to other methods	0.293	0.455	1,555
Drawback: difficult to learn to use	0.288	0.453	1,555
Drawback: fragile or prone to breaking	0.244	0.430	1,555
Drawback: hard to repair locally	0.102	0.303	1,555

2.3 Experimental Design and Treatment Arms

We implement a two-stage randomization design. First, villages are randomized into pure control or treatment. Pure control villages receive no intervention and allow us to measure spillover effects by comparing with control households in treated villages. Within treatment villages then we cross-randomize two interventions at the individual level: a three month long trial of the technology along with a short initial demonstration, and peer exposure through meal-sharing meetings with a randomly matched peer. Importantly, because peer matching is random, 47 percent of assigned pairs are cross-caste pairs.

Because peers are randomly matched and the technology access is randomized within each pair, we observe four pair compositions where neither peer has the technology, only the respondent has it, only the peer has it, and both have it. Half of the recipients of the

Table 5: Perceived Benefits, Demand and Electricity Access at Baseline

	Mean	SD	N
<i>Panel A. Beliefs and willingness to pay</i>			
Believed air-quality improvement from switching (percent)	68.0	22.4	1,555
Maximum willingness to pay for a cookstove (NPR)	3107.5	1317.2	1,511
Accepts the top price of NPR 5,000 (censored from above)	0.228	0.420	1,511
Rejects every price on the ladder	0.028	0.166	1,555
Willing to pay at least a tenth of own monthly income	0.784	0.411	1,511
<i>Panel B. Expected role of peers in adoption</i>			
Likely or very likely to adopt if a friend adopts	0.857	0.350	1,555
Reason: a friend would motivate me to use it	0.567	0.496	1,333
Reason: seeing a friend use it makes me confident it works	0.529	0.499	1,333
Reason: a friend adopting shows the cookstove is useful	0.440	0.497	1,333
Reason: a friend could teach me how to use it	0.292	0.455	1,333
Reason: a friend could share the cost	0.116	0.320	1,333
Of 10 closest friends, number thought to have used a cookstove	0.8	1.5	1,480
<i>Panel C. Electricity access</i>			
Frequent power cuts in the household	0.961	0.194	1,555
Monthly electricity bill (NPR)	222.1	441.0	1,551
Electricity connection capacity (amperes)	7.8	4.5	1,555

technology trial are additionally randomized to an air-quality demonstration: they are shown the reading from an air-quality monitor while cooking with firewood and again while cooking with the electric stove, aiming to demonstrate the reduction in indoor air pollution directly. Importantly, we exclude all individuals from the baseline who report having used a cookstove already from the experiment.

2.4 Meeting Structure

Peers are randomly matched and asked to meet every two weeks for three months. The first meeting is verified by an enumerator, who helps the pair set a meal plan and schedule future meetings. Subsequent meetings are unsupervised and incentivized, with a chance of a cash reward if both peers are present when the enumerator calls. When the technology is available to either or both, peers are invited to share a meal aiming to facilitate observational and joint learning by doing respectively. Pairs where neither have the technology are invited to meet to discuss any topic of their choice, aiming to facilitate information exchange.

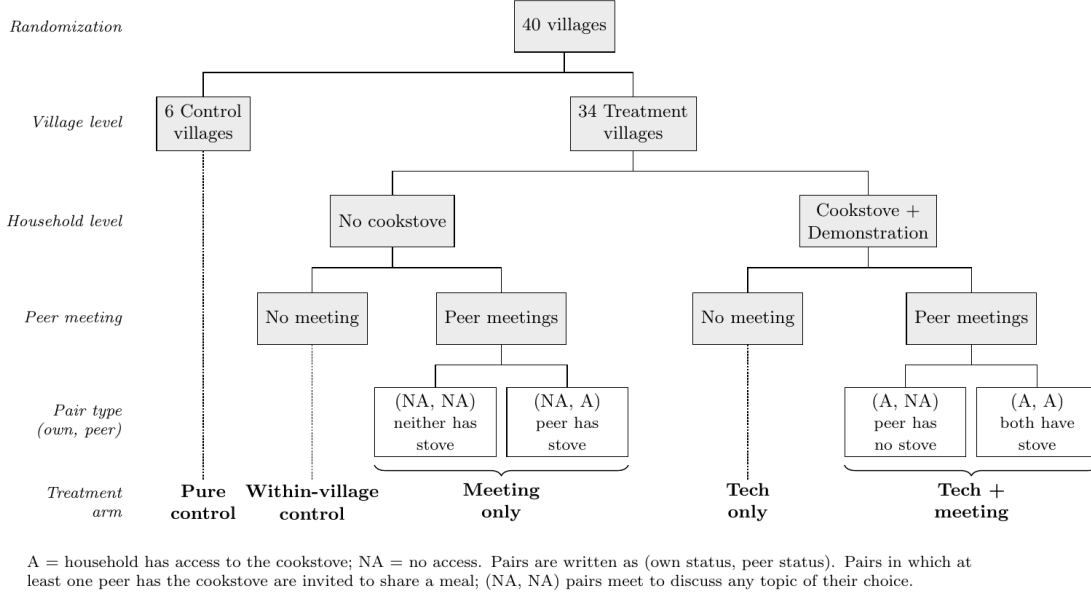


Figure 1: Experimental Design

2.5 Randomization and Balance

We check baseline balance for each of the three randomized assignments. Appendix A reports the tables. None of the 31 baseline covariates differs significantly between the meeting and no-meeting groups or between the technology-access and no-access groups. Among the 334 cookstove recipients, three covariates differ across the two air-quality reading groups at the 5 percent level: maximum willingness to pay, having completed primary schooling, and the number of friends named at baseline. We show robustness while studying the effect of the AQI reading using post-double-selection lasso (5).

2.6 Empirical Specification

We report results from three intent-to-treat (ITT) specifications, estimated using endline respondents in treatment villages, with robust standard errors.

The first specification compares the four arms. Tech_i is an indicator for technology access without a meeting, Meet_i an indicator for a meeting without the technology and TechMeet_i for both, with respondents assigned neither treatment included as the omitted category,

$$Y_i = \alpha + \beta_1 \text{Tech}_i + \beta_2 \text{Meet}_i + \beta_3 \text{TechMeet}_i + \varepsilon_i.$$

Because the three indicators are mutually exclusive cells, each coefficient shows the effect of that arm against the within-village control.

The second specification decomposes the meeting arm using own and peer ownership of the

technology. We construct Both_i , OwnOnly_i , PeerOnly_i and Neither_i as the four pair compositions depending on technology access, and compare against the same omitted category,

$$Y_i = \alpha + \beta_1 \text{Tech}_i + \beta_2 \text{Both}_i + \beta_3 \text{OwnOnly}_i + \beta_4 \text{PeerOnly}_i + \beta_5 \text{Neither}_i + \varepsilon_i.$$

We implement four additional tests, each isolating one channel of social learning. The test $\beta_4 = \beta_5$ isolates observational learning: both cells hold a meeting without own access, and they differ only in whether there is a technology to observe in the partner's hands. The test $\beta_3 = \beta_1$ isolates information exchange: respondents in both cells use the technology themselves, and the meeting adds a partner to talk to but nothing to observe, since she does not have the technology. The tests $\beta_2 = \beta_1$ and $\beta_2 = \beta_3$ isolate joint learning by doing: the first asks whether using the technology alongside an owning partner improves on using it alone, and the second asks, holding the meeting fixed, whether it matters that the partner also has the technology.

The third specification restricts attention to the meeting arm, where the outcomes are meeting-specific and only pairs that met can report them. Neither_i is the omitted category.

$$Y_i = \alpha + \beta_1 \text{Both}_i + \beta_2 \text{OwnOnly}_i + \beta_3 \text{PeerOnly}_i + \varepsilon_i.$$

3 Conceptual Framework

We model how individuals simultaneously choose how much to interact and learn from others. Woman i is assigned a peer j and chooses how much to interact with her, $m \geq 0$, and how much information to actively elicit from her, $q \in [0, m]$. Her utility from the relationship is

$$W_i(m, q) = b_{ij}m - \frac{c}{2}m^2 - \frac{\kappa_{ij}}{2}q^2 + H_i(r_i^0 + \alpha m + \beta q) - H_i(r_i^0) - F_{ij}\mathbf{1}\{m > 0\}. \quad (1)$$

The first term captures the direct benefit of interacting with the peer. The parameter b_{ij} is the relationship-specific marginal benefit of contact. Higher b_{ij} makes spending time with the peer more valuable independently of any information exchanged. Joint access to the technology can increase this benefit by giving the pair a shared activity. For instance, let $J_{ij} = 1$ when both i and j have access to the technology. Then, we can write

$$b_{ij} = \bar{b}_{ij} + \delta J_{ij}, \quad \delta > 0. \quad (2)$$

Thus, joint access directly increases the return to interaction.

The second term captures the cost of contact. We assume this cost is convex, with $c > 0$, so that increasingly intensive interaction becomes more costly.

The third term captures the cost of actively eliciting information. The woman chooses asking

effort $q \in [0, m]$, where $q \leq m$ captures the fact that asking requires interaction and therefore cannot be positive without an interaction. The parameter $\kappa_{ij} > 0$ is the cost of eliciting information. Joint access to a technology can increase this cost because, when both women have had an opportunity to learn how to use the technology, asking a question may reveal that one does not already know the answer. We therefore allow

$$\kappa_{ij} = \bar{\kappa}_{ij} + \eta J_{ij}, \quad \eta \geq 0. \quad (3)$$

Contact and elicitation both generate information. Let r_i^0 denote the precision of the information woman i possesses independently of the assigned peer, including information generated through her own experience with the technology. Information precision after interacting with the peer can be written as:

$$r_i(m, q) = r_i^0 + \alpha m + \beta q, \quad \alpha, \beta > 0. \quad (4)$$

Contact generates information passively at rate α , while asking questions generates additional ‘active’ information at rate β .

The final component of (1),

$$H_i(r_i^0 + \alpha m + \beta q) - H_i(r_i^0),$$

is the value of the additional information generated through the relationship. We now derive $H_i(r)$ from the woman’s technology-adoption decision.

The value of information. Woman i decides whether to adopt a technology of uncertain quality θ . With adoption $a_i \in \{0, 1\}$, her payoff is

$$u_i(a_i, \theta) = a_i(\theta - p_i), \quad (5)$$

where p_i is the cost of adopting the technology.

Initially,

$$\theta \sim \mathcal{N}(\mu_{i0}, \pi_{i0}^{-1}), \quad (6)$$

where μ_{i0} is expected quality and π_{i0} is prior precision. Suppose she receives an average signal x_i with precision r_i ,

$$x_i \mid \theta \sim \mathcal{N}(\theta, r_i^{-1}). \quad (7)$$

Bayes’ rule gives

$$\pi_{i1} = \pi_{i0} + r_i, \quad \mu_{i1} = \frac{\pi_{i0}\mu_{i0} + r_i x_i}{\pi_{i0} + r_i}. \quad (8)$$

After observing the signal, the woman adopts if and only if $\mu_{i1} \geq p_i$.

Before the signal is realized, her expected payoff from optimally choosing whether to adopt is therefore

$$H_i(r_i) = \mathbb{E}_{x_i} \left[\max \left\{ \frac{\pi_{i0}\mu_{i0} + r_i x_i}{\pi_{i0} + r_i} - p_i, 0 \right\} \right]. \quad (9)$$

We assume that H_i is twice continuously differentiable and that

$$H'_i(r) > 0. \quad (10)$$

Thus more precise information increases the expected payoff from the adoption decision.

3.1 Equilibrium and Comparative Statics

Returning to the relationship problem, conditional on activating the relationship, woman i chooses

$$(m^*, q^*) = \arg \max_{m \geq 0, 0 \leq q \leq m} \left\{ b_{ij}m - \frac{c}{2}m^2 - \frac{\kappa_{ij}}{2}q^2 + H_i(r_i^0 + \alpha m + \beta q) - H_i(r_i^0) \right\}. \quad (11)$$

She activates the relationship when the maximized value in (11) is at least F_{ij} . Otherwise $m^* = q^* = 0$. Consider an active relationship with an interior solution, $0 < q^* < m^*$. Equilibrium information precision is

$$r_i^* = r_i^0 + \alpha m^* + \beta q^*. \quad (12)$$

The first-order conditions are

$$b_{ij} - cm^* + \alpha H'_i(r_i^*) = 0, \quad (13)$$

$$-\kappa_{ij}q^* + \beta H'_i(r_i^*) = 0. \quad (14)$$

Equivalently,

$$m^* = \frac{b_{ij} + \alpha H'_i(r_i^*)}{c}, \quad (15)$$

$$q^* = \frac{\beta H'_i(r_i^*)}{\kappa_{ij}}. \quad (16)$$

Equation (15) shows that contact is higher when the relationship itself is more valuable and when the information generated through contact is more valuable. Since joint access increases b_{ij} , it creates a direct incentive toward greater interaction. Equation (16) shows that asking is lower when the cost of eliciting information is higher.

For the comparative statics, write $H'' = H''_i(r_i^*)$ and define

$$A = c - \alpha^2 H'', \quad D = \kappa_{ij} - \beta^2 H'', \quad B = -\alpha\beta H'', \quad \Delta = AD - B^2. \quad (17)$$

We assume the second-order conditions

$$A > 0, \quad D > 0, \quad \Delta > 0. \quad (18)$$

Relationship-specific benefits. We first ask how making the relationship itself more valuable affects interaction and information acquisition. This comparative static captures the direct channel through which a shared technology can increase engagement with the assigned peer.

Proposition 1. Relationship-specific benefits increase contact and information precision.

A higher relationship-specific benefit increases equilibrium contact and information precision:

$$\frac{\partial m^*}{\partial b_{ij}} = \frac{D}{\Delta} > 0, \quad \frac{\partial r_i^*}{\partial b_{ij}} = \frac{\alpha \kappa_{ij}}{\Delta} > 0. \quad (19)$$

It also increases the surplus from activating the relationship.

If information has diminishing marginal value, so that $H_i''(r_i^*) \leq 0$, a higher relationship-specific benefit can at the same time reduce active information elicitation:

$$\frac{\partial q^*}{\partial b_{ij}} = -\frac{B}{\Delta} \leq 0. \quad (20)$$

The reason is that a more valuable relationship induces greater contact, which itself generates passive information. With diminishing returns to information precision, this reduces the marginal value of actively eliciting additional information. Thus greater contact can increase total information precision even while reducing the amount of information actively elicited from the peer.

Because joint access increases b_{ij} , Proposition 1 implies that joint access increases contact. The shared technology creates a relationship-specific benefit that makes interaction worthwhile even independently of its informational value.

The cost of asking. Next, we ask how a higher cost of actively eliciting information affects behavior within an existing relationship. This isolates the learning friction that can arise when asking a question reveals that one does not already know the answer.

Proposition 2. The cost of asking reduces information elicitation and precision.

A higher asking cost reduces equilibrium asking and information precision:

$$\frac{\partial q^*}{\partial \kappa_{ij}} = -\frac{Aq^*}{\Delta} < 0, \quad \frac{\partial r_i^*}{\partial \kappa_{ij}} = -\frac{\beta c q^*}{\Delta} < 0. \quad (21)$$

If information has diminishing marginal value, it also weakly increases contact:

$$\frac{\partial m^*}{\partial \kappa_{ij}} = \frac{Bq^*}{\Delta} \geq 0. \quad (22)$$

As asking becomes more costly, active information acquisition falls. This increases the marginal value of the information obtained through contact, partially shifting information acquisition toward passive learning.

A higher asking cost also reduces the surplus from activating the relationship. Thus a higher cost of asking reduces information acquisition even within an active relationship.

3.2 Effect of Joint Access to the Technology

The previous results show how the two effects of joint access operate together. Joint access increases the relationship-specific benefit from $\bar{b}_{ij}$ to $\bar{b}_{ij} + \delta$ and increases the cost of asking from $\bar{\kappa}_{ij}$ to $\bar{\kappa}_{ij} + \eta$. Under diminishing marginal returns to information, both changes increase contact and reduce active information elicitation.

Proposition 3.

Let $m^*(J_{ij})$, $q^*(J_{ij})$, and $r_i^*(J_{ij})$ denote the equilibrium under joint-access status $J_{ij} \in \{0, 1\}$. Suppose $\delta > 0$, $\eta > 0$, both equilibria are active and interior, and $H_i''(r) \leq 0$ over the relevant range. Then joint access increases equilibrium contact and reduces active information elicitation:

$$m^*(1) > m^*(0), \quad q^*(1) < q^*(0). \quad (23)$$

The effect of joint access on total information precision is ambiguous since joint access both increases passive learning (due to higher interaction) and reduces information elicitation. The local comparative statics are

$$dr_i^* = \frac{\alpha \kappa_{ij}}{\Delta} db_{ij} - \frac{\beta c q^*}{\Delta} d\kappa_{ij}. \quad (24)$$

Thus, total information precision falls locally when

$$\frac{\delta}{\eta} < \frac{\beta c q^*}{\alpha \kappa_{ij}}. \quad (25)$$

Joint access can therefore increase interaction while reducing active information elicitation.

tion. Whether total information precision also falls depends on whether the increase in relationship-specific benefits, and hence passive information generated through additional contact, is large enough to offset the reduction in actively elicited information.

4 Effect on Technology Use

4.1 Adoption

We first measure adoption of the technology using five pre-specified outcomes including whether the respondent used the stove in the past three months, how often she used it, and for how many minutes a day as well as their monthly household electricity bill, and whether the household upgraded its wiring. Table 6 reports effects by treatment arm and Table 7 decomposes the meeting arm by pair composition.

Table 6: Adoption by Treatment Arm

	Used (3m)	Freq	Min/day	Elec. bill	Wiring
Technology only	0.662*** (0.038)	3.582*** (0.220)	21.540*** (1.846)	128.059*** (37.414)	0.085*** (0.030)
Meeting only	0.005 (0.010)	0.016 (0.050)	0.120 (0.338)	-14.333 (27.507)	0.026 (0.018)
Technology $\times$ meeting	0.769*** (0.032)	3.991*** (0.190)	23.273*** (1.631)	84.340** (36.364)	0.133*** (0.032)
Technology $\times$ meeting = Technology only	[0.028]	[0.153]	[0.478]	[0.323]	[0.232]
Technology $\times$ meeting = Meeting only	[0.000]	[0.000]	[0.000]	[0.007]	[0.001]
Control mean	0.016	0.083	0.517	230.323	0.054
Observations	1106	1106	1106	1104	1106
R^2	0.606	0.562	0.393	0.018	0.026

We find that the technology trial increases three-month use by 66 percentage points against a control mean of 1.6 percent, and also increases usage frequency, minutes of daily use, the monthly electricity bill (NPR 128, USD 0.8), and wiring upgrades (8 percentage points). Meeting, without technology access, has no significant effect on any of these outcomes. The combined arm exceeds technology alone only on self-reported use, by 11 percentage points ($p = 0.03$). This suggests that peer meetings had no incremental effect across most outcomes around adoption.

Table 7 further decomposes the meeting arm by pair composition. Pairs in which both partners have the technology report higher use than respondents who have it alone, by 17 percentage points relative to the technology-only arm ($p = 0.004$). They also report 10 points more use than pairs in which only the respondent has the technology, but that difference is

Table 7: Adoption by Pair Composition

	Used (3m)	Freq	Min/day	Elec. bill	Wiring
Technology only	0.662*** (0.038)	3.582*** (0.220)	21.540*** (1.848)	128.059*** (37.448)	0.085*** (0.030)
Meeting: neither has technology	-0.001 (0.009)	-0.008 (0.051)	-0.093 (0.312)	-27.556 (29.478)	0.020 (0.019)
Meeting: peer has technology	0.024 (0.020)	0.084 (0.097)	0.709 (0.774)	22.353 (44.284)	0.044 (0.032)
Meeting: respondent has technology	0.732*** (0.042)	3.944*** (0.245)	21.979*** (2.002)	79.163* (44.273)	0.108*** (0.037)
Meeting: both have technology	0.831*** (0.045)	4.071*** (0.295)	25.483*** (2.748)	93.318* (51.719)	0.177*** (0.054)
Meeting: both = Technology only	[0.004]	[0.180]	[0.232]	[0.546]	[0.123]
Meeting: peer = Meeting: neither	[0.222]	[0.348]	[0.299]	[0.272]	[0.476]
Meeting: respondent = Technology only	[0.205]	[0.265]	[0.871]	[0.337]	[0.608]
Meeting: both = Meeting: respondent	[0.107]	[0.739]	[0.301]	[0.820]	[0.277]
Control mean	0.016	0.083	0.517	230.323	0.054
Observations	1106	1106	1106	1104	1106
R^2	0.609	0.562	0.395	0.020	0.029

not statistically significant ($p = 0.11$). This difference appears only in self-reported use and not along other outcomes.

4.1.1 Willingness to Pay

Willingness to pay is elicited with an incentivized BDM mechanism on a price grid running from NPR 0 to NPR 5,000 (about USD 33). Figure 2 reports it by treatment arm and Figure 3 by pair composition.

We find that technology access significantly increases willingness to pay, while peer meetings do not have any incremental effect. Consistent with this, within the meeting arm, willingness to pay is higher when the respondent herself has access to the technology, regardless of whether her matched peer also has access.

4.2 Knowledge and Confidence

The limited effects on adoption are particularly surprising given how access to the technology increases engagement with the assigned peer. Joint-access pairs attend significantly more sessions, are much more likely to cook together, and are more likely to meet outside the scheduled sessions (Table 16). We first ask whether more interaction leads to higher

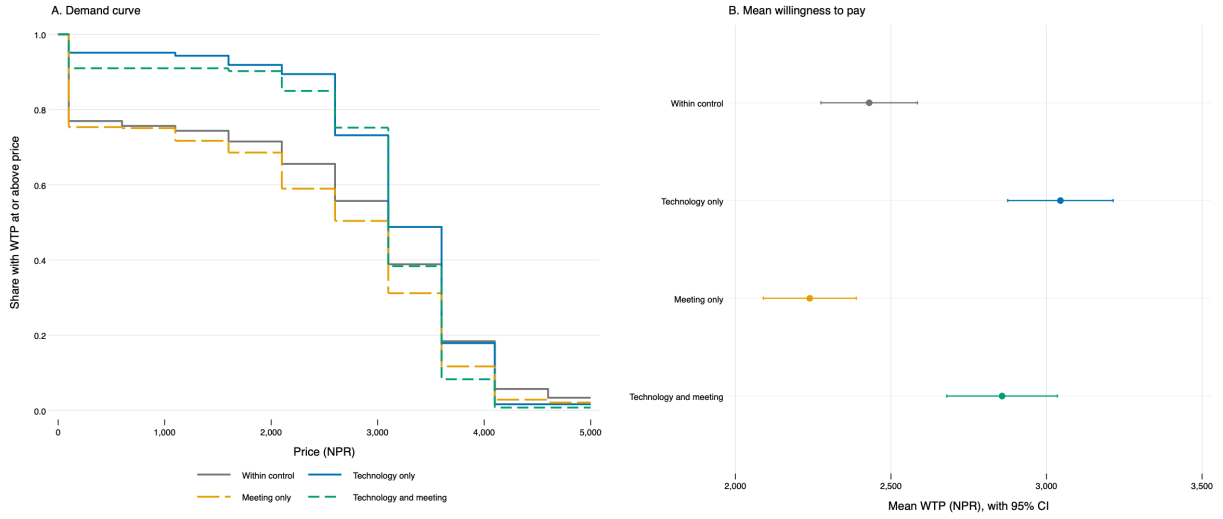


Figure 2: Willingness to Pay by Treatment Arm

Notes: Incentivized BDM willingness to pay, endline respondents in treatment villages. Panel A plots the share of the arm with a willingness to pay at or above each price on the NPR 0–5,000 grid. Panel B plots the arm mean with a 95 percent confidence interval.

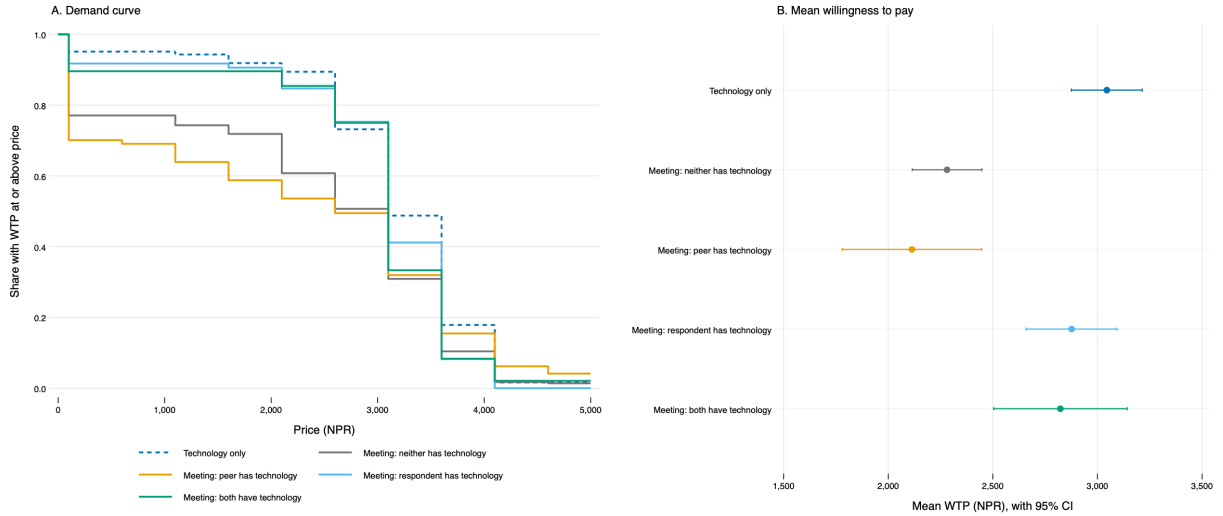


Figure 3: Willingness to Pay by Pair Composition

Notes: As in Figure 2, for the four pair compositions in the meeting arm, with the technology-only arm as the benchmark.

confidence or knowledge about how to use the technology.

We measure confidence using respondents’ reported confidence in operating the stove, coding those who report being “confident” or “very confident” as highly confident.

Table 8: Knowledge

	High confidence	Checklist
Technology only	0.547*** (0.041)	0.374*** (0.028)
Meeting: neither has technology	-0.033 (0.023)	-0.009 (0.018)
Meeting: peer has technology	0.046 (0.040)	0.035 (0.028)
Meeting: respondent has technology	0.583*** (0.047)	0.359*** (0.029)
Meeting: both have technology	0.674*** (0.054)	0.342*** (0.030)
Meeting: both = Technology only	[0.047]	[0.400]
Meeting: peer = Meeting: neither	[0.047]	[0.135]
Meeting: respondent = Technology only	[0.541]	[0.691]
Meeting: both = Meeting: respondent	[0.177]	[0.662]
Control mean	0.111	0.093
Observations	1104	1104
R^2	0.366	0.304

We find that joint access substantially increases confidence. Relative to the control group, the technology-only, respondent-only and joint-access cells are 55, 58 and 67 percentage points more likely to report high confidence, respectively (all $p < 0.01$). In particular, respondents in joint-access pairs are 13 percentage points more likely to report high confidence than respondents with technology access alone ($p = 0.047$).

However, this increase in confidence is not accompanied by a higher level of knowledge. We measure knowledge using an enumerator-administered operating checklist based on correct operating steps and unsafe recommendations. Relative to the control group, the technology-only, respondent-only and joint-access cells have checklist scores that are 0.37, 0.36 and 0.34 higher, respectively (all $p < 0.01$) with no differences across these groups. Knowledge is therefore not significantly higher among respondents with joint access to the technology than among those with technology access alone ($p = 0.400$). Thus, even though confidence is significantly higher under joint access, this does not translate into higher knowledge. If

anything, the estimated effect on knowledge is slightly smaller than in the technology-only group.

4.3 Attention and Exploration

Next, we study how the treatment affects respondents’ attention to different dimensions of the cooking activity and how concentrated their attention is on these dimensions. Table 9 reports these effects.

The first measure is breadth, which is the number of dimensions of cooking to which a respondent reports paying attention. We find no evidence that the treatment affects breadth. Respondents in the combined arm report paying attention to fewer dimensions than those with technology access alone, but the difference is not statistically significant ($p = 0.18$). Similarly, both-stove respondents report paying attention to fewer dimensions than technology-only respondents, but this difference is also not statistically significant ($p = 0.226$).

The next three measures capture how concentrated respondents’ attention is across different dimensions of cooking. We group the items respondents report paying attention to into five categories: sensory (taste or smell, smoke or heat), process (cooking time, ease or difficulty), resource (fuel or electricity cost), cleanliness and safety (cleanliness, safety, or children nearby), and social (how others cook or comment). Responses classified as “other” are excluded.

We construct three measures of concentration. The first is a Herfindahl index, $HHI_i = \sum_c p_{ic}^2$, where p_{ic} is the share of respondent i ’s selected items that fall in category c . It is higher when a respondent’s attention is concentrated in a small number of categories rather than spread across many of them. The second is pair share, $\sum_c \binom{n_{ic}}{2} / \binom{n_i}{2}$. This is the share of all possible pairs of selected items that come from the same category, and is therefore also higher when attention is concentrated.

The third measure is based on entropy, which is high when attention is spread relatively evenly across categories and low when it is concentrated in only a few. We convert entropy into a concentration measure by constructing the following:

$$1 - \frac{-\sum_c p_{ic} \log p_{ic}}{\log(\min\{n_i, 5\})}.$$

The denominator normalizes entropy by the maximum amount of dispersion possible given the number of items selected, up to the five available categories. The resulting measure is therefore higher when attention is more concentrated. All three measures are coded so that higher values indicate greater concentration. HHI is defined for the 1,100 respondents who select at least one categorized item, while pair share and entropy require at least two categorized items and are defined for 997 respondents.

Table 9: Attention Across Cooking Dimensions

	Breadth	HHI	Pair share	Entropy
<i>Panel A. Treatment arms</i>				
Technology only	0.135 (0.147)	-0.049** (0.020)	-0.099*** (0.021)	-0.105*** (0.024)
Meeting only	0.101 (0.097)	-0.016 (0.016)	-0.019 (0.021)	-0.019 (0.022)
Technology $\times$ meeting	-0.084 (0.118)	-0.007 (0.020)	-0.053** (0.023)	-0.052** (0.025)
Technology $\times$ meeting = Technology only	[0.180]	[0.069]	[0.050]	[0.047]
Technology $\times$ meeting = Meeting only	[0.126]	[0.649]	[0.135]	[0.191]
Control mean	3.010	0.557	0.253	0.302
Observations	1106	1100	997	997
R^2	0.003	0.005	0.017	0.016
<i>Panel B. Pair composition</i>				
Technology only	0.135 (0.147)	-0.049** (0.020)	-0.099*** (0.021)	-0.105*** (0.024)
Meeting: neither has technology	0.057 (0.104)	-0.017 (0.017)	-0.016 (0.023)	-0.018 (0.024)
Meeting: peer has technology	0.225 (0.168)	-0.011 (0.026)	-0.028 (0.030)	-0.021 (0.032)
Meeting: respondent has technology	-0.073 (0.146)	-0.009 (0.024)	-0.064** (0.026)	-0.064** (0.029)
Meeting: both have technology	-0.103 (0.160)	-0.003 (0.029)	-0.035 (0.034)	-0.031 (0.037)
Meeting: both = Technology only	[0.226]	[0.148]	[0.060]	[0.056]
Meeting: peer = Meeting: neither	[0.333]	[0.825]	[0.689]	[0.926]
Meeting: respondent = Technology only	[0.259]	[0.129]	[0.189]	[0.174]
Meeting: both = Meeting: respondent	[0.881]	[0.876]	[0.442]	[0.440]
Control mean	3.010	0.557	0.253	0.302
Observations	1106	1100	997	997
R^2	0.004	0.005	0.017	0.017

Notes: Breadth is the number of dimensions of cooking a respondent says she pays attention to. HHI, pair share, and entropy measure the concentration of attention across five categories: sensory, process, resource, cleanliness/safety, and social; “other” is excluded. HHI is the sum of squared category shares, pair share is the fraction of distinct pairs of selected items that belong to the same category, and entropy is one minus normalized entropy. Higher values on all three measures indicate more concentrated attention. Panel A is the arm-level specification and Panel B the pair-composition specification of Section 2.6.

However, we find a consistent pattern across all three measures of concentration. Panel A shows that technology access on its own spreads attention across categories: relative to the control group, HHI falls by 0.05, pair share by 0.10, and entropy concentration by 0.11. These reductions are smaller once a meeting is added. Relative to the technology-only

arm, concentration is higher in the combined arm under all three measures, with differences significant at the 10-percent level for HHI ($p = 0.069$) and at the 5-percent level for pair share ($p = 0.050$) and entropy ($p = 0.047$).

Panel B shows a similar pattern when both partners have the technology. Relative to technology-only respondents, both-stove respondents have more concentrated attention under all three measures. The difference is not statistically significant for HHI ($p = 0.148$), but is significant at the 10-percent level for pair share ($p = 0.060$) and entropy concentration ($p = 0.056$). Taken together, these measures consistently suggest that meeting with another technology user concentrates respondents' attention more heavily within a smaller set of categories.

4.4 Beliefs

We now examine how respondents' beliefs about the technology respond to the various treatments. Table 10 reports the effects.

The belief index is a weighted average of nine items capturing respondents' assessments of the stove's expected benefits and practical risks. We find that the technology trial increases the favorable-beliefs index by 0.35σ ($p < 0.01$), with significant gains on health, cooking time, suitability for traditional foods and concern about electric shocks, and the remaining items in the same direction but not individually significant. Meeting a peer without access does not affect the index, and the combined arm lies between the two at 0.18σ (Panel A).

Panel B decomposes the meeting arm. We find that pairs in which both partners have the technology have significantly less favorable beliefs, with an index 0.34σ below the technology-only arm ($p = 0.03$). Moreover, their beliefs are not statistically different from pairs in which neither partner has access. Thus, more interaction and higher confidence generated by joint access do not translate into more favorable assessments of the technology.

The pattern is different when we consider the case where the peer has experience with the technology but the respondent does not. Respondents who meet a peer with access but do not have the technology themselves have more favorable beliefs than pairs in which neither partner has access as per the overall index ($p = 0.063$). This provides evidence of observational learning from a peer's experience. Peer-access-only pairs are also the most likely to rank asking questions and getting help among the most helpful aspects of the meetings (Table 13). Taken together, these results suggest that learning from a peer who has experience with the technology is different from learning when both people have access to the technology.

Table 10: Beliefs

	Index	Health	Time	Trad. foods	AQI belief	Dal time	Elec. supply	Elec. bill	Breakage	Shocks
<i>Panel A. Treatment arms</i>										
Technology only	0.354*** (0.111)	0.078* (0.044)	0.138*** (0.051)	0.127** (0.057)	7.975 (6.199)	-0.092 (0.486)	0.030 (0.070)	0.087 (0.068)	0.094 (0.061)	0.177*** (0.065)
Meeting only	-0.061 (0.071)	0.075** (0.031)	0.005 (0.033)	0.072* (0.042)	2.322 (2.033)	0.730* (0.397)	-0.051 (0.042)	-0.134*** (0.043)	-0.057 (0.043)	-0.025 (0.040)
Technology $\times$ meeting	0.179* (0.092)	0.097** (0.041)	0.105** (0.044)	0.110** (0.050)	6.050** (2.914)	-0.345 (0.446)	-0.100* (0.054)	-0.024 (0.063)	-0.003 (0.059)	0.035 (0.055)
Technology $\times$ meeting = Technology only	[0.161]	[0.710]	[0.589]	[0.793]	[0.770]	[0.631]	[0.099]	[0.176]	[0.183]	[0.057]
Technology $\times$ meeting = Meeting only	[0.009]	[0.600]	[0.030]	[0.468]	[0.217]	[0.016]	[0.375]	[0.080]	[0.353]	[0.274]
Control mean	-0.000	3.180	3.124	3.009	80.506	14.561	2.048	2.063	2.132	1.907
Observations	1106	1077	1006	927	1106	1106	1071	1075	996	1067
R^2	0.019	0.008	0.014	0.008	0.005	0.006	0.005	0.014	0.007	0.013
<i>Panel B. Pair composition</i>										
Technology only	0.354*** (0.111)	0.078* (0.044)	0.138*** (0.051)	0.127** (0.057)	7.975 (6.204)	-0.092 (0.487)	0.030 (0.070)	0.087 (0.068)	0.094 (0.062)	0.177*** (0.065)
Meeting: neither has technology	-0.119 (0.075)	0.070** (0.034)	-0.007 (0.038)	0.046 (0.046)	1.561 (2.383)	0.810* (0.436)	-0.078* (0.045)	-0.142*** (0.047)	-0.069 (0.046)	-0.035 (0.043)
Meeting: peer has technology	0.098 (0.115)	0.090* (0.051)	0.039 (0.049)	0.147** (0.064)	4.435* (2.342)	0.508 (0.584)	0.022 (0.074)	-0.114 (0.070)	-0.024 (0.064)	0.002 (0.065)
Meeting: respondent has technology	0.274** (0.111)	0.150*** (0.050)	0.140** (0.055)	0.114* (0.060)	7.512** (3.532)	-0.471 (0.507)	-0.085 (0.066)	-0.009 (0.080)	0.046 (0.071)	0.074 (0.067)
Meeting: both have technology	0.016 (0.131)	0.007 (0.058)	0.048 (0.061)	0.102 (0.073)	3.555 (4.452)	-0.130 (0.662)	-0.126* (0.076)	-0.048 (0.083)	-0.084 (0.087)	-0.034 (0.078)
Meeting: both = Technology only	[0.030]	[0.286]	[0.219]	[0.765]	[0.551]	[0.958]	[0.103]	[0.169]	[0.066]	[0.024]
Meeting: peer = Meeting: neither	[0.063]	[0.711]	[0.396]	[0.139]	[0.299]	[0.620]	[0.197]	[0.701]	[0.499]	[0.587]
Meeting: respondent = Technology only	[0.566]	[0.225]	[0.972]	[0.863]	[0.946]	[0.513]	[0.191]	[0.316]	[0.563]	[0.222]
Meeting: both = Meeting: respondent	[0.099]	[0.042]	[0.225]	[0.884]	[0.461]	[0.642]	[0.659]	[0.718]	[0.206]	[0.251]
Meeting: both = Meeting: neither	[0.311]	[0.298]	[0.396]	[0.467]	[0.671]	[0.170]	[0.540]	[0.270]	[0.865]	[0.992]
Meeting: both = Meeting: peer	[0.603]	[0.247]	[0.902]	[0.610]	[0.851]	[0.418]	[0.134]	[0.508]	[0.547]	[0.701]
Control mean	-0.000	3.180	3.124	3.009	80.506	14.561	2.048	2.063	2.132	1.907
Observations	1106	1077	1006	927	1106	1106	1071	1075	996	1067
R^2	0.025	0.012	0.016	0.011	0.006	0.006	0.007	0.015	0.009	0.014

Notes: Column 1 is the belief index and columns 2–10 are the nine items it is built from: expected health benefits, time savings and suitability for traditional foods (agreement with a favourable statement, 1–4); the incentivized expected percentage improvement in indoor air quality; the expected number of minutes to cook dal on the stove; and concern about electricity reliability, the electricity bill, breakage and electric shocks (1–4, coded so that a higher value means less concern). Every column except “Dal time” is oriented so that a higher value is a more favourable belief; on “Dal time” the favourable direction is negative, and the item enters the index reverse-coded. Panel A is the arm-level specification and Panel B the pair-composition specification of Section 2.6.

4.5 Mechanisms

The absence of effects on joint-access pairs is particularly striking given that these pairs engage most with their assigned peer: they attend significantly more sessions, are much more likely to cook together, and are more likely to meet outside the scheduled sessions (Table 16).

4.5.1 Experiment 1: Information Elicitation Costs

The disclosure-cost mechanism makes a testable prediction. If beliefs in joint-access pairs are less favorable because asking a peer who also has access is embarrassing, then supplying the pair with information that is not embarrassing to elicit (as information provision is

randomized) should not involve these costs. The air-quality reading provides this variation: half of technology recipients, chosen at random, were shown the reading from an air-quality monitor while cooking with firewood and again while cooking with the electric stove.

Table 11 reports the result. We find that seeing the reading oneself has no effect on any outcome. Among joint-access pairs whose peer was *not* shown the reading, the belief index is 0.54σ lower than the technology-only arm ($p < 0.01$) and the knowledge index is 0.21σ lower, although the latter is not statistically significant. Among joint-access pairs whose peer *was* shown the reading, both gaps are statistically indistinguishable from zero (0.11σ below and 0.21σ above the technology-only arm). Importantly, the difference between the two joint-access cells (i.e. with and without the AQI reading) is 0.44σ on beliefs ($p = 0.07$) and 0.42σ on knowledge ($p = 0.085$). Moreover, we also find that the joint access pair where the peer saw the reading is 22 percentage points more likely to have discussed the technology ($p = 0.02$; Appendix B.2).

Table 11: The Peer’s Air-Quality Reading, Beliefs and Knowledge

	Belief Idx	Knowledge Idx	% AQI Improve	AQI Conf.
Both have technology, peer saw no reading	-0.544*** (0.158)	-0.212 (0.201)	-7.770 (7.957)	-0.119 (0.199)
Both have technology, peer saw the reading	-0.108 (0.230)	0.207 (0.200)	0.401 (8.678)	0.240 (0.158)
Peer saw the reading – peer saw no reading	0.436	0.419	8.171	0.359
<i>p</i> -value	[0.073]	[0.085]	[0.339]	[0.116]
Technology-only mean	0.394	1.607	94.310	4.184
Observations	223	223	223	223
R^2	0.028	0.010	0.011	0.011

Notes: Estimated on technology recipients who were either not assigned a meeting—the omitted category, which is the technology-only arm—or assigned a meeting with a peer who also received the technology. The 111 recipients whose assigned peer received no technology are excluded, so that the omitted category is the technology-only arm alone and the coefficients are comparable with the pair-composition estimates of Section 4. Robust standard errors in parentheses. The air-quality reading was randomized among technology recipients, independently for each member of a pair. Every regression controls for whether the respondent herself was shown the reading; that coefficient is not reported. The two joint-access rows are mutually exclusive levels rather than increments: each reports the difference from the technology-only arm for joint-access pairs whose peer was, or was not, shown the reading, and the row beneath them reports the difference between the two. The technology-only mean is taken over the omitted category among those who saw no reading themselves.

4.5.2 Experiment 2: Vignettes

We implement a survey experiment to understand the mechanism further, particularly focusing on the reluctance to ask questions among joint-access pairs, mentioned above. In a phone follow-up we asked meeting-arm respondents to predict, in hypothetical scenarios, how likely a person would be to ask a peer for help. The scenarios independently vary three attributes in a $2 \times 2 \times 2$ design: whether both people own the technology, whether the peer is a central

advisor in the village, and whether the peer is a close friend. Joint ownership increases expected embarrassment to ask questions by 0.10 points on the three-point scale ($p < 0.05$) and expected judgement by 0.16 points ($p < 0.01$), but has no statistically detectable effect on willingness to ask. Describing the peer as a close friend increases willingness to ask by 0.98σ ($p < 0.01$). Describing the peer as a central advisor reduces it ($p < 0.10$).

Table 12: Vignette Experiment: Willingness to Ask

	Willingness to ask	Not embarrassed	No judgement
Both own the technology	-0.034 (0.057)	-0.097** (0.038)	-0.163*** (0.040)
Peer is a central advisor	-0.086* (0.050)	-0.055 (0.036)	-0.014 (0.042)
Peer is a close friend	0.982*** (0.060)	0.557*** (0.040)	0.478*** (0.042)
Respondents	480	480	480
Observations	960	960	960
R^2	0.243	0.188	0.135

Notes: Respondent-by-scenario data from the phone follow-up: 480 respondents each evaluated two of eight scenarios, giving 960 observations. Scenario attributes are randomly assigned and independent. Outcomes are coded so that a higher value is the more favourable response. Standard errors are clustered by respondent. Willingness to ask is the standardized average of two items: how likely the character in the scenario is to ask her peer for help, and whether the respondent would ask if she were in the character’s place.

These vignette estimates are consistent with an elicitation cost: joint access increases expected embarrassment and judgement to ask questions.

4.5.3 Additional Evidence

Asking Questions in the Meetings Respondents in the meeting arm were asked to rank the three things they found most helpful about the meetings. Table 13 reports, by pair composition, the share ranking each aspect in the top three.

Pairs in which both partners have the technology are 17 percentage points less likely than pairs with no technology to rank “asking questions and getting help” among the three most helpful aspects ($p < 0.10$), and 25 points less likely than pairs in which only the peer has it ($p = 0.03$).

Stated Barriers to Asking Table 14 reports how respondents describe the barriers to asking, first about themselves and then about people in general.

Responses to the third-person questions identify prevalent barriers: 62 percent state that

Table 13: What Peers Found Helpful in the Meetings

	Fun/time	Easier learn	Trusted	Ask help	Noticed	Safety	Motivated	Reminded
Meeting: peer has technology	0.021 (0.097)	0.026 (0.104)	0.016 (0.109)	0.082 (0.109)	-0.127* (0.076)	0.220** (0.102)	-0.136*** (0.045)	-0.070 (0.051)
Meeting: respondent has technology	0.115 (0.088)	0.120 (0.097)	0.141 (0.110)	-0.075 (0.102)	-0.064 (0.085)	-0.061 (0.078)	0.052 (0.083)	-0.102** (0.040)
Meeting: both have technology	0.049 (0.100)	0.080 (0.106)	0.092 (0.117)	-0.171* (0.099)	0.039 (0.101)	-0.001 (0.092)	-0.062 (0.068)	-0.065 (0.054)
Meeting: neither mean	0.729	0.661	0.390	0.356	0.220	0.186	0.136	0.102
Meeting: respondent = Meeting: peer	[0.356]	[0.401]	[0.321]	[0.195]	[0.455]	[0.009]	[0.008]	[0.318]
Meeting: both = Meeting: peer	[0.805]	[0.656]	[0.568]	[0.032]	[0.101]	[0.059]	[0.149]	[0.905]
Meeting: both = Meeting: respondent	[0.527]	[0.721]	[0.707]	[0.386]	[0.339]	[0.533]	[0.192]	[0.316]
Observations	150	150	150	150	150	150	150	150
R^2	0.011	0.011	0.013	0.032	0.023	0.058	0.046	0.033

Table 14: Comfort Asking the Peer and Barriers to Asking

	Mean	SD	N
<i>Panel A. Why a woman in the village might not ask her partner</i>			
She would feel she should already know the answer	0.25	0.44	226
She would not want to seem as if she did not know how to use it	0.62	0.49	226
She would think her partner would not know either	0.73	0.45	226
The conversation would have moved on	0.17	0.38	226
She would not feel comfortable enough with her partner	0.07	0.26	226
<i>Panel B. The respondent's own experience</i>			
Comfort asking her peer a question (1 = very, 4 = not at all)	1.74	0.79	232
Ever wanted to ask her peer something and did not	0.03	0.16	232
<i>Among those who held back, why she did not ask</i>			
I felt I should already know the answer	0.00	0.00	6
I did not want to seem as if I did not know how to use it	0.17	0.41	6
I thought my partner would not know either	0.83	0.41	6
The conversation moved on	0.33	0.52	6
I was not comfortable enough with my partner	0.00	0.00	6

Notes:

she might not want to appear ignorant. Twenty-five percent state that she might believe she should already know the answer. At the same time, 73 percent state that someone might not ask because she expects her partner not to know. This response supports shared ignorance as a possible constraint on learning in addition to elicitation costs.

Prior Relationships We now explore whether prior acquaintance reduces elicitation costs. We define prior acquaintance as women at social distance one, meaning that they are directly connected to the peer. Panel A of Table 15 restricts attention to the cell in which the

mechanism should operate: women assigned both the technology and a meeting, whose peer therefore also has access. The omitted category is the 33 such women who did not know their peer and the reported coefficient is for the 115 who did.

Among these women, those who knew their peer use the technology more intensively: use frequency is 0.96 points higher (4.93 versus 3.97; $p = 0.023$) and daily use 6.0 minutes longer ($p = 0.08$), and they are 11 percentage points more likely to know how to operate the stove (0.96 versus 0.85), though the last of these is imprecise ($p = 0.10$). Whether they used the stove at all does not differ, and neither willingness to pay nor the belief index shows a detectable difference. Across the whole meeting arm (Panel B), women who knew their peer attended 0.9 more sessions and were 20 percentage points more likely both to cook with the peer and to meet her outside the scheduled sessions (all $p < 0.01$). The pattern on engagement and use is consistent with the vignette result that describing the peer as a close friend increases willingness to ask by 0.98σ , while the absence of an effect on beliefs indicates that familiarity raises contact and use more clearly than it shifts assessments of the technology.

Table 15: Meeting Effects by Prior Relationship with the Peer

	Used (3m)	Freq	Min/day	Knows oper.	WTP	Beliefs
<i>Panel A. Technology $\times$ meeting only; omitted category is women who did not know their peer</i>						
Knew peer	-0.005 (0.057)	0.961** (0.419)	6.029* (3.439)	0.108 (0.066)	134.655 (229.903)	0.125 (0.163)
Mean, did not know peer	0.909	3.970	22.727	0.848	2810.345	0.134
Observations	148	148	148	148	129	148
R^2	0.000	0.039	0.015	0.032	0.004	0.003
<i>Panel B. Whole meeting arm, controlling for pair stove composition</i>						
	Sessions	Cooked together	Met outside	Knows oper.	WTP	Beliefs
Knew peer	0.915*** (0.264)	0.201*** (0.066)	0.195*** (0.057)	0.080** (0.040)	-105.123 (169.527)	-0.028 (0.104)
Mean, did not know peer	1.410	0.147	0.277	0.384	2646.341	0.084
Observations	371	204	371	381	363	382
R^2	0.088	0.323	0.047	0.593	0.059	0.032

Notes: Prior acquaintance is reported at the intervention rollout, when the match was announced, and is observed for 406 of the 610 meeting-arm women. Panel A is estimated on the 148 women assigned both the technology and a meeting whose relationship with the peer is reported; the omitted category is those who did not know her, and the row reports the single difference for those who did. Panel B is estimated on the whole meeting arm, controlling for pair stove composition, so that the engagement outcomes defined only inside the meeting arm can be reported. Robust standard errors in parentheses. Willingness to pay is the incentivized BDM measure in NPR and the belief index is defined as in Table 10.

5 Effect on Social Interactions

5.1 Peer Take-up and Engagement

We first check whether peer meeting assignment translated into actual meetings. We measure engagement with the number of sessions attended out of six, whether the pair cooked together, whether they met outside the scheduled sessions, the change in their meeting frequency, and the amount donated to the partner in a dictator game. Table 16 reports these by pair composition (Panel A) and by caste (Panel B).

Table 16: Engagement in the Meeting Arm

	Sessions	Cooked together	Met outside	Frequency vs 3 months ago	Donation
<i>Panel A. Pair composition</i>					
Meeting: peer has technology	0.267 (0.271)	0.266*** (0.067)	0.039 (0.057)	0.042 (0.068)	51.339 (38.536)
Meeting: respondent has technology	0.226 (0.262)	0.494*** (0.071)	0.080 (0.055)	0.199*** (0.065)	-40.722 (33.142)
Meeting: both have technology	1.813*** (0.341)	0.531*** (0.076)	0.190*** (0.070)	0.204*** (0.076)	32.264 (42.896)
Meeting: both = Meeting: peer	[0.000]	[0.008]	[0.062]	[0.071]	[0.706]
Meeting: both = Meeting: respondent	[0.000]	[0.715]	[0.166]	[0.954]	[0.118]
Meeting: neither mean	1.531	0.034	0.335	1.927	389.145
Observations	546	265	546	546	546
R^2	0.053	0.265	0.015	0.024	0.009
<i>Panel B. Caste composition</i>					
Cross-caste pair	-0.649*** (0.206)	-0.122** (0.054)	-0.198*** (0.043)	-0.089* (0.053)	-61.642** (28.845)
Meeting: both have technology	0.666 (0.415)	0.273** (0.110)	-0.038 (0.088)	0.032 (0.077)	-6.871 (57.474)
Cross-caste $\times$ meeting: both have technology	2.438*** (0.622)	0.172 (0.157)	0.462*** (0.132)	0.273* (0.156)	82.975 (79.682)
Cross-caste gap, no joint access (β_1)	-0.649 [0.002]	-0.122 [0.025]	-0.198 [0.000]	-0.089 [0.094]	-61.642 [0.033]
Cross-caste gap, joint access ($\beta_1 + \beta_3$)	1.789 [0.002]	0.049 [0.738]	0.263 [0.035]	0.184 [0.210]	21.333 [0.774]
Joint access, cross-caste pairs ($\beta_2 + \beta_3$)	3.104 [0.000]	0.445 [0.000]	0.423 [0.000]	0.305 [0.025]	76.104 [0.168]
Same-caste, not both mean	1.945	0.268	0.455	2.024	419.537
Observations	546	265	546	546	546
R^2	0.083	0.104	0.056	0.015	0.010

Notes: Meeting arm only. Panel A uses the engagement specification of Section 2.6, with pairs where neither partner has the technology as the omitted category. Panel B replaces pair composition with a cross-caste indicator, an indicator for both partners having the technology, and their interaction. The three rows below the coefficients report the cross-caste gap without joint access (β_1), the cross-caste gap with joint access ($\beta_1 + \beta_3$), and the effect of joint access among cross-caste pairs ($\beta_2 + \beta_3$); p -values in brackets. Cooking together is asked only of respondents who attended at least one session, which is why that column has fewer observations.

Panel A reports engagement by pair composition. Pairs in which both partners have the technology attend 1.8 more of the six sessions than pairs in which neither does ($p < 0.01$), are 53 percentage points more likely to have cooked together, and are 19 points more likely to have met outside the scheduled sessions. Pairs in which only the respondent has the technology attend no more often than pairs with no technology, which suggests that it is joint access rather than own access that leads to sustained attendance. Donations are large

in every cell, averaging NPR 389 (USD 2.5) out of an endowment of NPR 1,000 (USD 6.5) in pairs where neither has the technology, and do not differ by composition.

Panel B reports engagement by caste composition of the pair. We find that cross-caste pairs engage less than same-caste pairs: they attend 0.65 fewer sessions ($p = 0.002$), are 12 percentage points less likely to have cooked together, and are 20 points less likely to have met outside the sessions. Joint access reverses this pattern along two margins. Among cross-caste pairs, joint access raises attendance by 3.1 of the six sessions ($p < 0.001$) and the probability of meeting outside the sessions by 43 percentage points ($p < 0.001$). As a result, the cross-caste gap not only closes but reverses on attendance: with joint access, cross-caste pairs attend 1.8 more sessions than same-caste pairs at the same level of access ($p = 0.002$). This suggests that a shared activity can bridge gaps across caste boundaries. We will return to this in Section 5.2.

Table 17: Session Topics

	Breadth	Use	Foods	Problems	Safety	Benefits	vs FW/LPG	Experience	Future use
Meeting: peer has technology	0.369 (0.294)	0.149* (0.085)	0.118 (0.082)	-0.028 (0.061)	0.156** (0.076)	0.025 (0.068)	0.229*** (0.079)	0.013 (0.046)	-0.091 (0.061)
Meeting: respondent has technology	0.109 (0.195)	0.296*** (0.078)	0.030 (0.077)	0.074 (0.069)	-0.039 (0.059)	-0.029 (0.061)	0.066 (0.069)	0.007 (0.043)	0.013 (0.069)
Meeting: both have technology	0.588* (0.304)	0.446*** (0.070)	0.144* (0.085)	0.133* (0.077)	0.046 (0.071)	0.060 (0.073)	0.103 (0.076)	0.062 (0.055)	-0.062 (0.066)
Meeting: neither mean	2.325	0.402	0.291	0.171	0.171	0.179	0.179	0.068	0.214
Meeting: respondent = Meeting: peer	[0.379]	[0.126]	[0.362]	[0.190]	[0.019]	[0.487]	[0.079]	[0.909]	[0.165]
Meeting: both = Meeting: peer	[0.561]	[0.001]	[0.795]	[0.058]	[0.232]	[0.684]	[0.198]	[0.444]	[0.677]
Meeting: both = Meeting: respondent	[0.117]	[0.072]	[0.246]	[0.515]	[0.270]	[0.274]	[0.677]	[0.376]	[0.347]
Observations	265	265	265	265	265	265	265	265	265
R^2	0.021	0.119	0.016	0.020	0.027	0.005	0.037	0.007	0.010

Table 17 reports what the pairs discussed during the meetings. Joint-access pairs discuss how to use the technology more than pairs with no technology (0.45, $p < 0.01$) and discuss foods and cooking problems more as well. Pairs in which only the peer owns the technology instead emphasize the comparison with firewood and LPG (0.23, $p < 0.01$) and safety. Discussion of technology-relevant content was therefore substantial, which suggests that limited engagement does not explain the belief and knowledge results of Section 4.

5.2 Cross-Caste Interaction

Next, we find that joint access to the technology also narrows the cross-caste gap in interaction. Because pairs are matched at random and caste is not used in matching, 47 percent of pairs are cross-caste. Panel B of Table 16 showed that cross-caste pairs attend 0.65 fewer sessions and are 20 percentage points less likely to meet outside the scheduled sessions, and that both of these gaps close, and on attendance reverse, when both partners have the technology. A shared activity can bridge gaps in a setting where cross-group interaction is rare and difficult to engineer directly.

5.3 Loneliness

Next, Table 18 reports effects on the UCLA loneliness index and its components.

Table 18: Loneliness

	Index	Companionship	Left out	Isolated
<i>Panel A. Treatment arms</i>				
Technology only	-0.060 (0.091)	-0.035 (0.046)	-0.009 (0.044)	-0.026 (0.042)
Meeting only	0.079 (0.071)	0.036 (0.036)	0.030 (0.034)	0.033 (0.033)
Technology $\times$ meeting	-0.114 (0.089)	-0.086** (0.044)	-0.076* (0.040)	0.003 (0.042)
Technology $\times$ meeting = Technology only	[0.602]	[0.335]	[0.179]	[0.548]
Technology $\times$ meeting = Meeting only	[0.028]	[0.005]	[0.010]	[0.486]
Control mean	-0.000	0.421	0.326	0.292
Observations	1106	1106	1106	1106
R^2	0.005	0.007	0.006	0.002
<i>Panel B. Pair composition</i>				
Technology only	-0.060 (0.091)	-0.035 (0.046)	-0.009 (0.044)	-0.026 (0.042)
Meeting: neither has technology	0.109 (0.077)	0.035 (0.039)	0.042 (0.037)	0.058 (0.037)
Meeting: peer has technology	-0.003 (0.106)	0.040 (0.056)	-0.002 (0.052)	-0.037 (0.049)
Meeting: respondent has technology	-0.005 (0.108)	-0.016 (0.053)	-0.064 (0.048)	0.050 (0.051)
Meeting: both have technology	-0.302** (0.122)	-0.206*** (0.057)	-0.095* (0.058)	-0.077 (0.056)
Meeting: both = Technology only	[0.072]	[0.008]	[0.182]	[0.417]
Meeting: peer = Meeting: neither	[0.309]	[0.932]	[0.421]	[0.067]
Meeting: respondent = Technology only	[0.652]	[0.750]	[0.324]	[0.182]
Meeting: both = Meeting: respondent	[0.042]	[0.006]	[0.649]	[0.063]
Control mean	-0.000	0.421	0.326	0.292
Observations	1106	1106	1106	1106
R^2	0.009	0.013	0.006	0.008

Notes: Panel A is the arm-level specification and Panel B the pair-composition specification of Section 2.6, on the three-item UCLA loneliness scale, coded so that a higher value means more lonely. Components are indicators for reporting the feeling sometimes or often.

Panel A shows that individuals in the technology access and peer exposure arm have significantly lower measures of loneliness: the treatment reduces lack of companionship by 0.09 ($p < 0.05$) and feeling left out by 0.08 ($p < 0.10$). Panel B shows where the effect is concentrated. Pairs where both partners have technology access report 0.30σ less loneli-

ness ($p < 0.05$), driven by a 21 percentage point fall in lacking companionship ($p < 0.01$). No other cell drives the effects on loneliness. Joint-access pairs are significantly less lonely than both the technology-only arm ($p = 0.07$) and pairs where only the respondent has the technology ($p = 0.04$).

5.4 Social Networks

We measure social interaction at the endline using outcomes including: whether the respondent often visits or is visited by another household, whether she often shares a meal or tea with someone, the number of people she names as friends, the number she exchanges advice with, and the number she borrows from or lends to. The index is a standardized inverse-covariance weighted combination of these outcomes. The three counts come from the endline network roster: respondents list everyone in the village they consider a friend, and are then asked with which of those friends they borrowed or lent money or goods, and with which they discussed or sought advice about personal or family problems, in the last month. House visits and shared meals are indicators for doing each often or very often over the same month. Table 19 reports the arm-level estimates (Panel A) and the pair-composition decomposition (Panel B).

Panel A reports the arm-level estimates. Neither treatment increases network interactions on its own. Peer exposure without technology access in fact lowers it. The index decreases by 0.17σ ($p < 0.05$), with 0.29 fewer advice contacts ($p < 0.01$) and fewer house visits. The combined meeting and technology access arm does not increase interaction above baseline levels, but it is significantly higher than the meeting-only arm ($p = 0.04$). It is also significantly higher than the control group on borrowing and lending, with respondents reporting 0.20 more people they borrow from or lend to ($p < 0.10$).

Panel B reports the pair-composition decomposition. The reduction in network interactions is concentrated in pairs where neither partner has the technology. Pairs in which both partners have it are indistinguishable from control.

5.5 Persistence: Six-Month Follow-up

Six months after the intervention, contact persists and remains highest where both partners have the technology. We asked how often respondents were still in touch with their assigned peer, on a four-point frequency running from never to weekly or more often. Table 20 reports effects by pair composition within the meeting arm.

Joint-access pairs report 0.32 more on the contact scale than pairs where neither has the technology ($p = 0.05$), against a mean of 2.75 for the latter, which corresponds to being in touch a few times a month. The effect sits at the top of the scale: joint-access pairs are 22 percentage points more likely to be in touch weekly or more often ($p < 0.01$) against a base

Table 19: Network Interaction

	Index	House visits	Shared meals	Friends	Advice	Borrow/lend
<i>Panel A. Treatment arms</i>						
Technology only	-0.137 (0.095)	0.005 (0.038)	-0.044 (0.045)	0.004 (0.087)	-0.303** (0.119)	-0.074 (0.119)
Meeting only	-0.174** (0.073)	-0.050* (0.030)	-0.020 (0.034)	-0.025 (0.064)	-0.290*** (0.089)	-0.106 (0.085)
Technology $\times$ meeting	0.015 (0.091)	0.008 (0.036)	-0.018 (0.043)	0.099 (0.080)	-0.122 (0.117)	0.200* (0.111)
Technology $\times$ meeting = Technology only	[0.170]	[0.945]	[0.618]	[0.336]	[0.197]	[0.050]
Technology $\times$ meeting = Meeting only	[0.042]	[0.114]	[0.964]	[0.121]	[0.148]	[0.006]
Control mean	-0.000	0.798	0.677	3.623	2.760	2.788
Observations	1106	1106	1106	1106	1037	1035
R^2	0.007	0.004	0.001	0.002	0.012	0.008
<i>Panel B. Pair composition</i>						
Technology only	-0.137 (0.095)	0.005 (0.038)	-0.044 (0.045)	0.004 (0.087)	-0.303** (0.119)	-0.074 (0.119)
Meeting: neither has technology	-0.188** (0.079)	-0.060* (0.033)	-0.023 (0.037)	-0.018 (0.069)	-0.290*** (0.098)	-0.128 (0.094)
Meeting: peer has technology	-0.134 (0.118)	-0.024 (0.046)	-0.010 (0.053)	-0.044 (0.104)	-0.290** (0.129)	-0.048 (0.127)
Meeting: respondent has technology	0.055 (0.104)	0.030 (0.041)	0.008 (0.050)	0.089 (0.090)	-0.112 (0.136)	0.250** (0.126)
Meeting: both have technology	-0.054 (0.145)	-0.029 (0.056)	-0.062 (0.065)	0.116 (0.128)	-0.141 (0.182)	0.109 (0.179)
Meeting: both = Technology only	[0.601]	[0.573]	[0.807]	[0.427]	[0.413]	[0.357]
Meeting: peer = Meeting: neither	[0.659]	[0.463]	[0.813]	[0.810]	[0.995]	[0.552]
Meeting: respondent = Technology only	[0.112]	[0.601]	[0.377]	[0.425]	[0.222]	[0.033]
Meeting: both = Meeting: respondent	[0.501]	[0.348]	[0.355]	[0.851]	[0.888]	[0.483]
Control mean	-0.000	0.798	0.677	3.623	2.760	2.788
Observations	1106	1106	1106	1106	1037	1035
R^2	0.008	0.005	0.002	0.002	0.012	0.009

of 17 percent, and pairs in which only the peer has the technology are 16 points more likely ($p < 0.10$). What joint access sustains is the intensity of contact rather than whether any contact survives, since 93 percent of the omitted group report some contact six months on.

When we asked what that contact involved, it was overwhelmingly social. Among those still in touch, 96 percent describe it as casual socializing and about half as sharing news (48 percent) or personal matters (48 percent), while only 11 percent describe it as giving or receiving advice. Giving or receiving advice is also the one that moves most across compositions, from 8 percent where neither partner has the technology to 18 percent where both do, and 16 percent report having contacted the peer for advice overall.

Table 20: Six-Month Follow-up

	Contact 1–4	Weekly or more often
Meeting: peer has technology	0.203 (0.160)	0.159* (0.082)
Meeting: respondent has technology	0.172 (0.140)	0.088 (0.073)
Meeting: both have technology	0.323* (0.164)	0.224*** (0.086)
Meeting: both = Meeting: respondent	[0.399]	[0.171]
Meeting: neither mean	2.750	0.167
Observations	231	231
R^2	0.021	0.039

Notes: Six-month phone follow-up, meeting arm only, with pairs where neither partner has the technology as the omitted category. Robust standard errors in parentheses. Column 1 is the four-point contact scale running from 1, never, to 4, weekly or more often; column 2 is an indicator for the top category.

A Randomization and Balance

Table 21: Baseline Balance: Peer Meeting Assignment

	No meeting	Meeting	<i>p</i> -value
Monthly income (NPR, winsorized 1 percent)	23112.219	22450.996	0.563
Age (years)	44.004	43.763	0.765
Household size	4.029	4.059	0.784
Total cookstoves ever used	7.492	7.633	0.760
Frequent power cuts	0.961	0.961	0.953
Primary fuel: firewood	0.523	0.517	0.842
Primary fuel: gas	0.473	0.483	0.748
Primary fuel: other	0.004	0.004	0.977
Believed cookstove cooking time (minutes)	16.393	17.005	0.452
Believed firewood cooking time (minutes)	23.745	24.029	0.579
Believed cookstove electricity bill (NPR)	356.886	354.640	0.917
Believed AQI improvement from cookstove (percent)	68.523	68.253	0.845
Max willingness to pay (NPR)	3179.775	3064.457	0.150
WTP zero indicator	0.020	0.032	0.216
Education: none	0.161	0.176	0.506
Education: informal	0.196	0.187	0.699
Education: primary	0.127	0.143	0.436
Education: grades 6–10	0.327	0.326	0.989
Education: grades 11–12	0.134	0.119	0.468
Education: university	0.055	0.048	0.603
Caste: Tamang	0.543	0.538	0.873
Caste: Chettri	0.240	0.226	0.583
Caste: Newar	0.075	0.064	0.470
Caste: Brahmin	0.046	0.053	0.561
Caste: Dalit	0.055	0.078	0.120
Caste: other	0.033	0.032	0.930
Caste: Rai	0.007	0.007	0.967
Network degree: house visits	2.304	2.289	0.786
Network degree: non-family visits	1.798	1.822	0.719
Network degree: friends	3.096	3.047	0.263
Network degree: non-family friends	2.463	2.513	0.465

Notes: Each row reports the mean by assignment and the *p*-value from a regression of the covariate on meeting assignment.

Table 22: Baseline Balance: Cookstove Assignment

	No cookstove	Cookstove	<i>p</i> -value
Monthly income (NPR, winsorized 1 percent)	22632.967	23109.336	0.702
Age (years)	43.652	44.413	0.385
Household size	4.052	4.027	0.832
Total cookstoves ever used	7.415	7.907	0.327
Frequent power cuts	0.961	0.961	0.996
Primary fuel: firewood	0.525	0.509	0.633
Primary fuel: gas	0.473	0.491	0.578
Primary fuel: other	0.005	0.000	0.188
Believed cookstove cooking time (minutes)	16.510	17.150	0.471
Believed firewood cooking time (minutes)	23.903	23.856	0.933
Believed cookstove electricity bill (NPR)	355.881	355.437	0.985
Believed AQI improvement from cookstove (percent)	68.556	67.994	0.709
Max willingness to pay (NPR)	3107.333	3154.434	0.589
WTP zero indicator	0.028	0.021	0.472
Education: none	0.168	0.171	0.927
Education: informal	0.187	0.204	0.508
Education: primary	0.140	0.123	0.444
Education: grades 6–10	0.329	0.320	0.778
Education: grades 11–12	0.126	0.129	0.887
Education: university	0.051	0.054	0.816
Caste: Tamang	0.553	0.512	0.208
Caste: Chettri	0.232	0.237	0.866
Caste: Newar	0.070	0.069	0.948
Caste: Brahmin	0.045	0.060	0.307
Caste: Dalit	0.063	0.075	0.487
Caste: other	0.030	0.039	0.433
Caste: Rai	0.006	0.009	0.652
Network degree: house visits	2.287	2.319	0.611
Network degree: non-family visits	1.791	1.854	0.393
Network degree: friends	3.071	3.071	0.995
Network degree: non-family friends	2.484	2.500	0.824

Table 23: Baseline Balance: AQI Reading Assignment (Cookstove Recipients)

	No AQI reading	AQI reading	<i>p</i> -value
Monthly income (NPR, winsorized 1 percent)	22245.562	23973.107	0.428
Age (years)	43.892	44.934	0.472
Household size	4.060	3.994	0.734
Total cookstoves ever used	8.060	7.754	0.723
Frequent power cuts	0.958	0.964	0.778
Primary fuel: firewood	0.509	0.509	1.000
Primary fuel: gas	0.491	0.491	1.000
Primary fuel: other	0.000	0.000	.
Believed cookstove cooking time (minutes)	17.701	16.599	0.560
Believed firewood cooking time (minutes)	23.952	23.760	0.829
Believed cookstove electricity bill (NPR)	343.671	367.204	0.494
Believed AQI improvement from cookstove (percent)	65.892	70.096	0.091
Max willingness to pay (NPR)	3311.729	3000.000	0.031
WTP zero indicator	0.030	0.012	0.253
Education: none	0.138	0.204	0.110
Education: informal	0.222	0.186	0.416
Education: primary	0.186	0.060	0.000
Education: grades 6–10	0.281	0.359	0.128
Education: grades 11–12	0.138	0.120	0.625
Education: university	0.036	0.072	0.147
Caste: Tamang	0.533	0.491	0.445
Caste: Chettri	0.204	0.269	0.158
Caste: Newar	0.060	0.078	0.518
Caste: Brahmin	0.060	0.060	1.000
Caste: Dalit	0.090	0.060	0.300
Caste: other	0.042	0.036	0.778
Caste: Rai	0.012	0.006	0.563
Network degree: house visits	2.248	2.389	0.193
Network degree: non-family visits	1.839	1.870	0.808
Network degree: friends	2.994	3.148	0.046
Network degree: non-family friends	2.463	2.537	0.559

B Endline Results

B.1 Willingness to Pay

Table 24: Willingness to Pay: Regression Estimates

	Treatment arms	Pair composition
Technology only	614.7*** (116.3)	
Meeting only	-191.1* (109.6)	
Technology $\times$ meeting	427.1*** (119.5)	
Technology only		614.7*** (116.4)
Meeting: neither has technology		-148.8 (115.8)
Meeting: peer has technology		-316.6* (185.0)
Meeting: respondent has technology		446.4*** (134.4)
Meeting: both have technology		392.9** (176.1)
Technology $\times$ meeting = Technology only	[0.130]	
Technology $\times$ meeting = Meeting only	[0.000]	
Meeting: both = Technology only		[0.216]
Meeting: peer = Meeting: neither		[0.371]
Meeting: respondent = Technology only		[0.224]
Meeting: both = Meeting: respondent		[0.780]
Control mean	2430.1	2430.1
Observations	1027	1027
R^2	0.039	0.040

Notes: Incentivized BDM willingness to pay in NPR, endline respondents in treatment villages, robust standard errors in parentheses. Column 1 is the arm-level specification and column 2 the pair-composition specification of Section 2.6, with respondents assigned neither treatment as the omitted category.

Table 25: Willingness to Pay: Kolmogorov–Smirnov Tests

	D	p
<i>Panel A. Treatment arms</i>		
Within control vs. Technology only	0.239	0.000
Within control vs. Meeting only	0.077	0.180
Within control vs. Technology $\times$ meeting	0.195	0.001
Technology only vs. Meeting only	0.305	0.000
Technology only vs. Technology $\times$ meeting	0.104	0.432
Meeting only vs. Technology $\times$ meeting	0.260	0.000
<i>Panel B. Pair composition</i>		
Technology only vs. Meeting: neither has technology	0.287	0.000
Technology only vs. Meeting: peer has technology	0.358	0.000
Technology only vs. Meeting: respondent has technology	0.097	0.682
Technology only vs. Meeting: both have technology	0.154	0.312
Meeting: neither has technology vs. Meeting: peer has technology	0.131	0.132
Meeting: neither has technology vs. Meeting: respondent has technology	0.246	0.000
Meeting: neither has technology vs. Meeting: both have technology	0.247	0.008
Meeting: peer has technology vs. Meeting: respondent has technology	0.318	0.000
Meeting: peer has technology vs. Meeting: both have technology	0.318	0.002
Meeting: respondent has technology vs. Meeting: both have technology	0.078	0.986

Notes: Two-sample Kolmogorov–Smirnov tests of the equality of the willingness-to-pay distributions plotted in Figures 2 and 3. D is the largest absolute difference between the two empirical distribution functions and p is the exact two-sided p -value. Panel A covers every pair of treatment arms and Panel B every pair of cells in the pair-composition decomposition.

B.2 Air-Quality Readings

Table 26: Own AQI Reading: Adoption and WTP

	Has CS	Used (3m)	Usage Idx	WTP New	WTP Used
AQI reading	-0.036 (0.039)	-0.006 (0.049)	-0.200 (0.269)	18.857 (158.063)	-104.126 (78.444)
Control Mean (no AQI)	0.868	0.737	1.623	869.697	284.848
Observations	334	334	334	331	331
R^2	0.003	0.000	0.002	0.000	0.005

Table 27: Own AQI Reading: Beliefs and Knowledge

	Belief Idx	% AQI Improve	AQI Conf.	Health Belief	Knowledge Idx	Know Operate
AQI reading	0.063 (0.124)	-5.665 (6.350)	-0.000 (0.103)	-0.024 (0.052)	0.092 (0.134)	0.030 (0.044)
Control Mean (no AQI)	0.230	90.299	4.269	3.280	1.627	0.778
Observations	334	334	334	328	334	334
R^2	0.001	0.002	0.000	0.001	0.001	0.001

Table 28: Own AQI Reading: Health and Network

	Health Idx	Resp. Symp.	Eye Irrit.	Headache	Network Idx	House Visits	Discuss CS	Lonely Idx
AQI reading	-0.033 (0.096)	-0.054 (0.043)	0.072 (0.056)	0.054 (0.058)	0.095 (0.110)	-0.006 (0.043)	0.048 (0.050)	-0.033 (0.105)
Control Mean (no AQI)	0.195	0.216	0.263	0.323	-0.104	0.808	0.671	-0.072
Observations	334	334	334	334	334	334	334	334
R^2	0.000	0.005	0.005	0.003	0.002	0.000	0.003	0.000

Table 29: Own vs Peer AQI Reading: Adoption and WTP

	Has CS	Used (3m)	Usage Idx	WTP New	WTP Used
Saw the reading herself	-0.018 (0.048)	-0.021 (0.060)	-0.379 (0.325)	-94.069 (197.087)	-189.405* (102.047)
Both have technology, peer saw no reading	0.024 (0.068)	0.176** (0.072)	-0.391 (0.440)	-94.066 (277.658)	-77.366 (123.047)
Both have technology, peer saw the reading	0.140*** (0.044)	0.163** (0.076)	0.462 (0.464)	-218.473 (264.021)	107.254 (150.148)
Peer saw the reading — peer saw no reading	0.116 [0.095]	-0.013 [0.883]	0.853 [0.139]	-124.407 [0.719]	184.620 [0.287]
p -value					
Technology-only mean	0.828	0.678	1.799	1114.943	413.793
Observations	223	223	223	221	221
R^2	0.018	0.030	0.015	0.004	0.017

Table 30: Own vs Peer AQI Reading: Health and Network

	Health Idx	Resp. Symp.	Eye Irrit.	Headache	Network Idx	House Visits	Discuss CS	Lonely Idx
Saw the reading herself	-0.084 (0.121)	-0.047 (0.053)	0.123* (0.071)	0.071 (0.073)	-0.041 (0.138)	-0.051 (0.054)	0.070 (0.061)	0.034 (0.126)
Both have technology, peer saw no reading	0.247 (0.167)	-0.116* (0.065)	-0.073 (0.103)	-0.133 (0.093)	-0.086 (0.192)	-0.126 (0.087)	0.029 (0.089)	-0.370** (0.156)
Both have technology, peer saw the reading	0.089 (0.153)	-0.102 (0.070)	-0.086 (0.088)	0.080 (0.111)	0.271 (0.220)	0.071 (0.068)	0.253*** (0.066)	-0.105 (0.190)
Peer saw the reading – peer saw no reading	-0.158	0.014	-0.012	0.213	0.357	0.197	0.224	0.264
<i>p</i> -value	[0.436]	[0.871]	[0.920]	[0.108]	[0.185]	[0.051]	[0.022]	[0.236]
Technology-only mean	0.106	0.264	0.310	0.345	-0.121	0.816	0.609	0.004
Observations	223	223	223	223	223	223	223	223
<i>R</i> ²	0.012	0.019	0.018	0.017	0.010	0.022	0.042	0.020

Table 31: Own AQI Reading by Baseline Air-Quality Beliefs: Beliefs and Knowledge

	Belief Idx	% AQI Improve	AQI Conf.	Health Belief	Knowledge Idx	Know Operate
AQI (at mean BL belief)	0.033 (0.125)	-6.587 (6.819)	-0.008 (0.104)	-0.023 (0.052)	0.075 (0.133)	0.023 (0.045)
Baseline belief (pp)	0.008* (0.004)	0.372 (0.283)	0.000 (0.003)	-0.001 (0.002)	0.002 (0.005)	0.002 (0.002)
AQI × baseline belief	-0.001 (0.006)	-0.305 (0.298)	0.004 (0.005)	0.001 (0.002)	0.004 (0.006)	-0.001 (0.002)
Control Mean (no AQI)	0.230	90.299	4.269	3.280	1.627	0.778
Observations	334	334	334	328	334	334
<i>R</i> ²	0.021	0.013	0.004	0.002	0.009	0.010

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